UNIT 5 · 9 hours

555 Timer IC in Astable Mode



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Describe the internal blocks of the 555 timer
- Explain astable (free-running) operation
- Derive/use the frequency and duty cycle equations
- Design an astable multivibrator

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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555 timer: internal blocks

Voltage divider

Three equal 5 kΩ resistors set reference levels 2Vcc/3 and Vcc/3.

Two comparators

Threshold comparator (2Vcc/3) and trigger comparator (Vcc/3) drive the flip-flop.

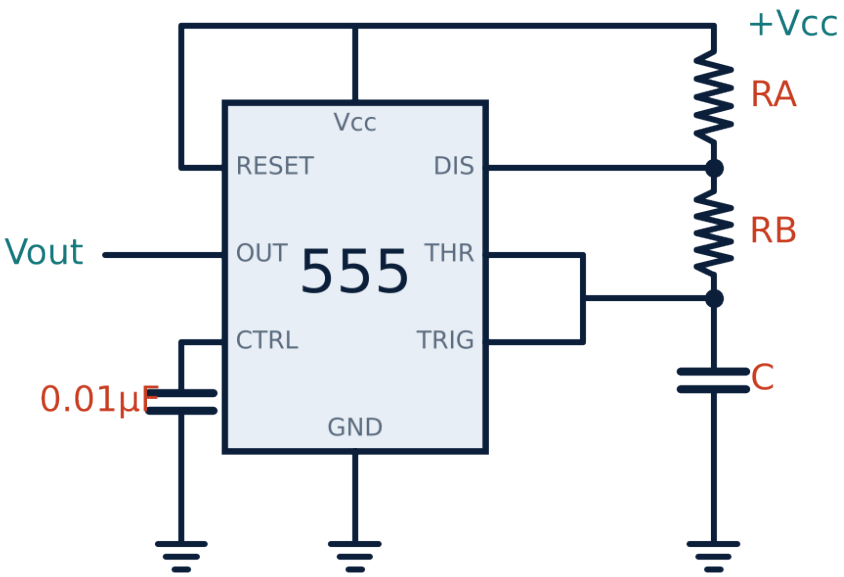
Flip-flop and discharge

Flip-flop sets the output; the discharge transistor discharges the timing capacitor.

8-pin IC: 1 GND, 2 TRIG, 3 OUT, 4 RESET, 5 CTRL, 6 THR, 7 DIS, 8 Vcc.

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Astable circuit



- External: RA, RB, C.
- Pins 2 and 6 tied together across C.
- Pin 7 (discharge) between RA and RB.
- Pin 4 to Vcc; pin 5 decoupled with 0.01 µF.
- No external trigger needed: free-running.

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How it oscillates

1

Charge

Output HIGH; C charges through RA + RB toward Vcc.

2

Reach 2Vcc/3

Threshold comparator resets the flip-flop; output LOW; discharge transistor ON.

3

Discharge

C discharges through RB toward ground.

4

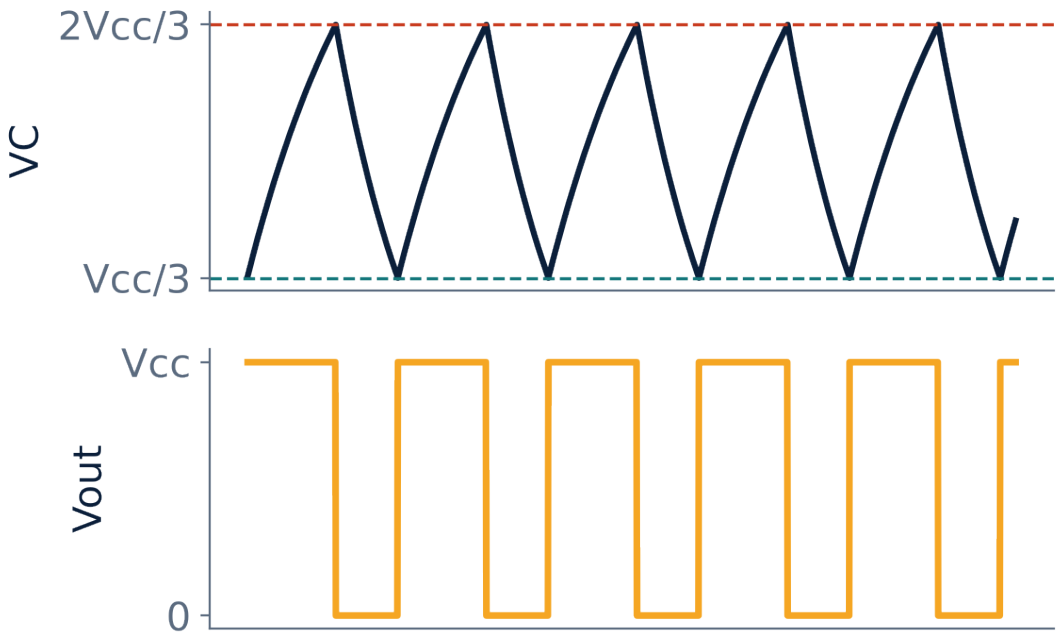
Reach Vcc/3

Trigger comparator sets the flip-flop; output HIGH again. Repeats.

$T_{HIGH} = 0.693\ (R_A + R_B)\ C$ **$T_{LOW} = 0.693\ R_B\ C$**

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Capacitor voltage and output waveform



- VC rises and falls between Vcc/3 and 2Vcc/3.
- Output is a rectangular wave.
- HIGH time is always longer than LOW time (RA > 0).

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Frequency and duty cycle

$$T = 0.693\ (RA + 2RB)\ C$$

period

$$f = 1.44\ /\ [(RA + 2RB)\ C]$$

frequency

$$D = (RA + RB)\,/\,(RA + 2RB)$$

duty cycle (> 50 %)

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Numerical: astable design check

Problem RA = 1 kΩ, RB = 10 kΩ, C = 0.1 µF. Find THIGH, TLOW, frequency and duty cycle.

- THIGH = 0.693 × 11 kΩ × 0.1 µF = 0.762 ms
- TLOW = 0.693 × 10 kΩ × 0.1 µF = 0.693 ms
- T = 1.455 ms → f = 1.44/(21 kΩ × 0.1 µF) = 686 Hz
- D = 11/21 = 52.4 %

Answer: f ≈ 687 Hz, duty cycle 52.4 %

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Key takeaways

- 555: two comparators, SR flip-flop, discharge transistor, 3 × 5 kΩ divider
- Astable: C charges to 2Vcc/3 and discharges to Vcc/3
- THIGH = 0.693(RA + RB)C; TLOW = 0.693 RB C
- f = 1.44/[(RA + 2RB)C]; D = (RA + RB)/(RA + 2RB)

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Practice questions

Q1. Draw and explain the 555 astable multivibrator.

Q2. Find f and duty cycle for RA = 2.2 kΩ, RB = 4.7 kΩ, C = 10 nF.

Q3. How would you get an approximately 50 % duty cycle?

Q4. State the pin functions of the 555.